Manuscript outline

Cell enlargement, polarity competition and migration persistence

Vision-aligned manuscript outline with proposed main figures and detailed legends

19 October 2026 · Current Figures 2–4 candidates for author discussion

Scientific argument and manuscript plan

Working hypothesis: WGD-associated enlargement changes how established polarity is organized and replaced; these changes may provide a physical basis for greater migration persistence. The theoretical contribution tests size under unchanged local kinetic parameters. Length differs between model cells to represent WGD-associated enlargement, rather than elongation during a turn. Existing migration observations supply empirical motivation; new paired WGD motility results retain a substantial, explicitly reserved contribution.

Figures 2–4 focus on three questions: which measurements separate explanations for difficult polarity replacement; how local activity becomes global polarity on a ring; and how physical site motion, pattern organization and geometry contribute to angular memory. Retained simulations and targeted distance–duration and Spring-ring experiments supply the evidence. Figure 2 now uses the separate finite-response candidate and Figure 3 the completed three-size stationary-cue review. Their emphasis and main composition remain provisional. Detailed assay maps and secondary diagnostics are supporting material.

This is a working manuscript outline. Figures 1 and 5 and the reserved Figure 6 experimental section retain their preceding content. Activity polarity, modelled replacement, observed migration and conditional OU velocity memory remain distinct readouts; their biological connection is an experimental hypothesis.

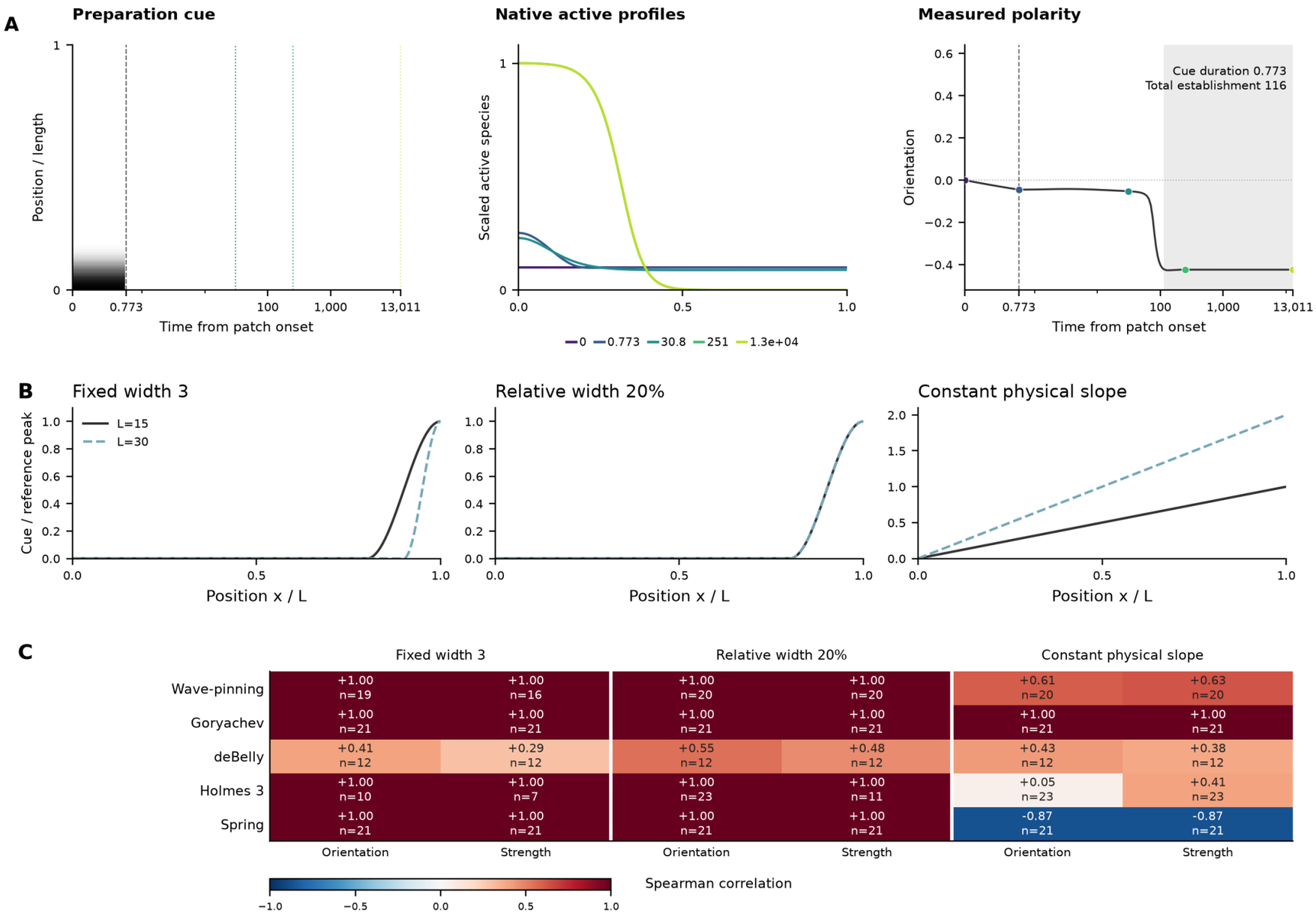
Proposed main figures

- Figure 1. Enlargement changes the cue duration required for lasting polarity replacement
- Figure 2. Distinct finite responses to local polarity challenges
- Figure 3. Local activity and retained orientation under stationary cues
- Figure 4 · Directional memory and physical site dynamics

Manuscript outline · continued

- Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation
- Figure 6. RESERVED: new WGD motility-assay results

Figure 1. Five models and three finite challenges



Current canonical comparison, 10 October Spearman revision; corrected independent-endpoint deBelly.

Figure 1. Enlargement changes the cue duration required for lasting polarity replacement

Comparative theoretical foundation: five established models, three finite challenges, and two distinct scalar replacement gates.

Detailed legend

A, Example wave-pinning preparation from a homogeneous state using a finite left patch. The cue ends at 0.773 native time units, whereas the subsequent basal relaxation establishes the polarized profile later (reported establishment arrival about 116). Coloured profiles and orientation samples distinguish input duration from autonomous development. The grey late interval is verification, not additional stimulation. Profiles are scaled for display.

B, Three rightward finite challenges restart the identical qualifying nominal-amplitude, relative-20% basal checkpoint at each model and length: fixed physical support 3 (Spring 0.2 native length units), relative support 20%, and a linear ramp with constant physical slope. Patch peaks remain fixed within each model; a fixed-slope ramp has a larger peak in a longer domain. Goryachev patches use the independently selected, length-independent multiplier K=64 (peak 6.4), which makes this a response-permissive comparison rather than a calibrated equal biochemical dose across models.

C, Spearman rank correlation between length and the shortest tested passing pulse duration, separately for orientation replacement and strength replacement, across Wave-pinning, Goryachev, corrected deBelly, Holmes 3 and Spring. Each cell reports the correlation and included successful-length count. All audited scalar successes enter, including those failing the separate native-field tail criterion. At least four distinct successful lengths and nonconstant durations are required; censored and preparation-ineligible cases receive no invented time. The companion duration curves retain all lengths and field-status markers.

After each positive pulse, input returns to the native basal value for 10,000 native time units. Orientation requires all 201 final-500 samples to have scalar orientation above 0.05 and contrast at least 0.05, with orientation range below 0.005. Strength additionally requires every tail orientation to be within 1% of the absolute initial leftward scalar orientation; it does not compare individual pole heights or amounts. The separate native-field tail flag requires pointwise range and projected 500-unit drift within 0.1% of frozen field scales. The reported duration excludes preparation and recovery. It is the shortest passing tested duration, with sampled fail/pass boundaries refined to 1%; narrower unobserved intervals are not excluded.

deBelly uses independently commanded corrected endpoints, rather than the deprecated in-loop centered extension. Spring length and time retain native units and are not silently converted to micrometres or seconds. Model-specific axes and stimulus definitions prevent interpreting cross-model amplitudes or clocks as biologically calibrated equivalents.

Figure 1. Enlargement changes the cue duration required for lasting polarity replacement · continued

Narrative points and conclusions

- Fixed-width and relative-width patches show perfectly increasing rank order among observed successes in Wave-pinning, Goryachev, Holmes 3 and Spring; enlargement often increases the finite stimulation required for a retained response.
- The response depends on the cue geometry and model. Spring has a negative constant-slope association (Spearman correlation -0.874 for both gates), while Holmes orientation under the constant-slope cue is nearly unassociated with length (0.047).
- Corrected deBelly curves are nonmonotone despite moderate positive rank correlations. The full duration curves and eligibility ranges are essential to the result.
- These are deterministic polarity-replacement experiments. They motivate, but do not measure, spontaneous polarity memory or migration persistence.

Proposed measurements and experimental follow-up

Mechanism-linked cellular follow-up would match local input support, amplitude and duration, track the original and challenged activity regions through stimulation and withdrawal, and distinguish orientation replacement from recovered activity strength. This validation is proposed and has no experimental results in this outline.

Scope, limitations and pending validation

Correlations are conditional on observed scalar success, rather than an unbiased summary of all possible sizes or a universal scaling law. Ineligible starts are distinct from responses unobserved by the finite horizon.

A retained scalar response can coexist with a changing native field. Only 13 of the 72 corrected deBelly scalar successes pass the separate native-field tail flag. Resolution checks from the earlier centered-deBelly campaign cannot qualify the corrected replacement thresholds.

Sources and reproducibility

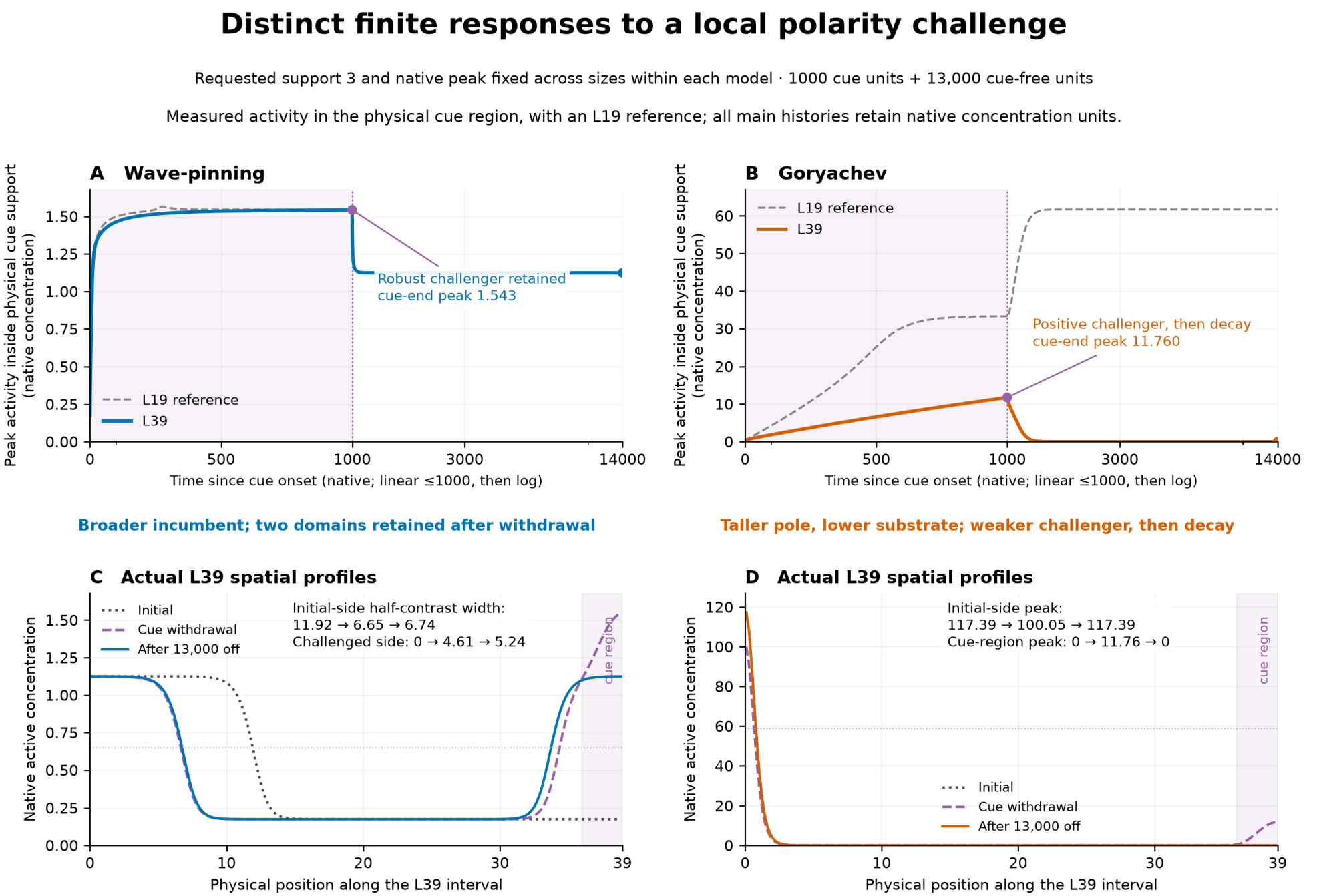
Three-test protocol and Spearman rule

Correlation table

Corrected endpoint method

Current duration curves

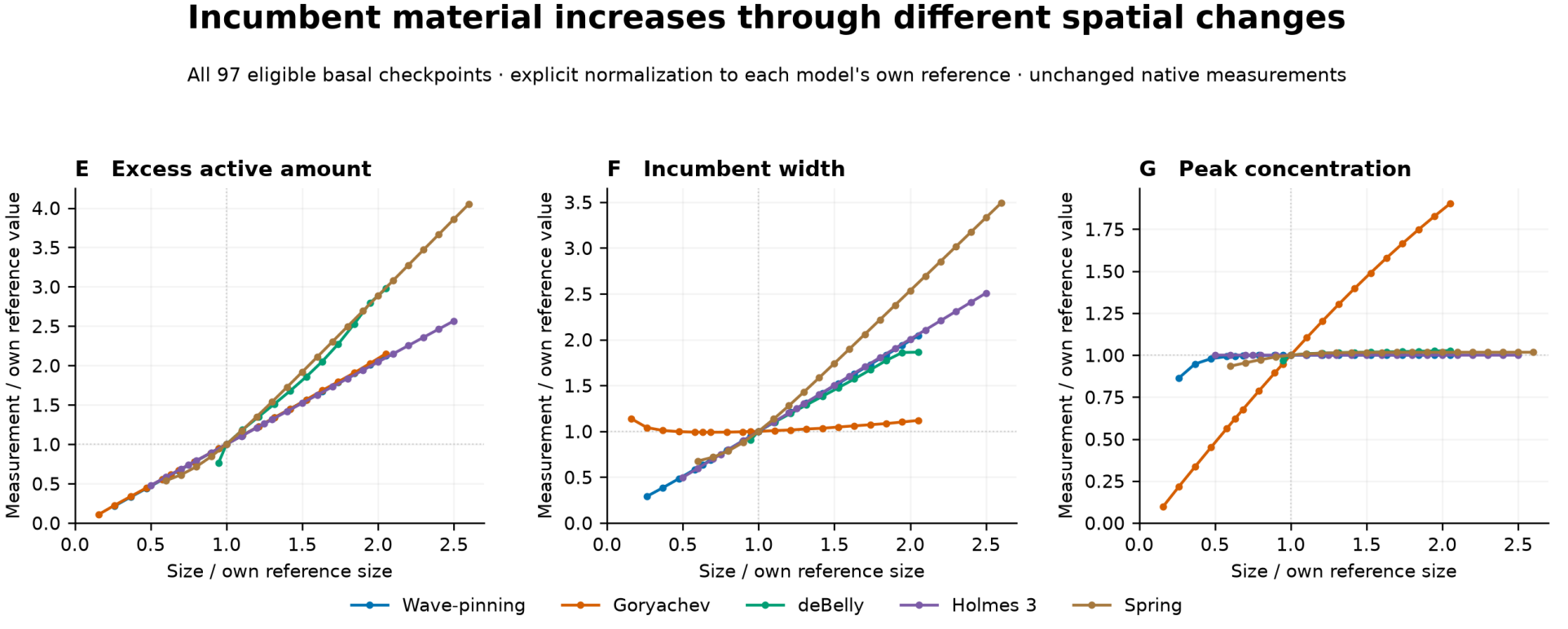
Figure 2. Finite challenge responses · image 1 of 2



Both native inputs transfer stimulus × inactive into active; finite responses include coupled reaction and transport.
Native peaks: WP 0.1; Goryachev 6.4. These are not matched optical doses or biological clocks.
Widths use each initial half-contrast threshold; zero threshold-defined extent does not mean absent activity. Finite patterns are not tracked site identities or a movement law.

Finite native activation and withdrawal responses

Figure 2 continued. Incumbent organization · image 2 of 2



All-five incumbent organization

Figure 2. Distinct finite responses to local polarity challenges

Similar size-dependent reversal behavior can arise through distinct mechanisms, which predict different observable responses to polarity challenges.

Detailed legend

A–B: Actual peak activity in the physical right-hand cue footprint [L−3,L] at L19 (gray reference) and L39 (model colour). Each model uses the same requested physical support 3, peak and 1000-unit duration across those sizes: WP 0.1, Goryachev 6.4, followed by 13000 cue-free native units. Both inputs add stimulus×inactive to active, with equal loss from inactive. Time is linear through 1000 then logarithmic, with withdrawal marked at 1000; the final coordinate 14000 is 13000 after withdrawal. Concentrations are native and are never independently peak-normalized. Goryachev L39 develops a positive peak 11.7598; it is 10.581 at 10 off and 5.2796 at 100 off, then approximately zero by 1000 off. Its L19 reference instead grows to 61.676 after an initial small withdrawal dip. WP's two sizes have nearly equal challenger height, including cue-end peaks 1.5460/1.5434, despite different remaining incumbents.

C–D: Exact L39 spatial fields initially, at withdrawal, and after 13000 cue-free units. Purple shading is the physical cue footprint, not activity extent. Dotted horizontal lines give each case's initial background-plus-half-range threshold. WP39 retains two connected above-threshold domains in every saved off-frame (2027 samples); their final initial-side/challenged-side widths are 6.7382/5.2372. Goryachev L39 never forms a challenger above its initial half-contrast 58.695, but its positive cue-driven activity is explicitly visible in B and in the withdrawal profile. Its initial-side peak recovers from 100.052 at withdrawal to 117.390. Profile values are measured fields, not evidence inferred from a scalar-strength gate. The endpoint two-field comparisons in the source table remain distinct from any scalar criterion. Connected regions refer to spatial patterns, not tracked identities or protrusions.

E–G: All 97 retained basal checkpoints, with excess amount, connected half-contrast width and peak concentration divided by the value at each model's own reference size. Sizes use L/Lref (WP/Goryachev/deBelly Lref 19; Holmes 20; Spring resting L_0ref 1). Amount is the one-dimensional excess-concentration integral within the basal half-contrast core, not calibrated molecule number. Spring uses its actual physical domain for widths/amounts; deBelly retains nominal chemical-solver coordinates. Thirteen ineligible preparations remain in the linked table, without artificial zero values. At L19→39, WP's excess amount/width/peak folds are 2.124/2.042/1.000, versus Goryachev 2.152/1.119/1.903. All five curves and the original full native-unit organization artwork remain accessible.

The labels linking organization to response name conditional model predictions, not unique molecular identification or a causal allocation of reversal time. The native channels, doses, spatial discretization and clocks are not optical-actuator or reporter calibrations. The chemical grid has 201 cells at both sizes; the sampled input peak is 0.999388/0.997422 of the requested peak at L19/L39 (about 0.2% mismatch), while its

Figure 2. Distinct finite responses to local polarity challenges · continued

physical integral is nearly unchanged. This small cue-sampling mismatch does not bound basal or other discretization errors; Goryachev L39's initial core has four sample centres and spatial convergence is not established by the retained checks. Thirteen thousand units of off-follow-up do not prove an asymptotic attractor. No velocity law or cell displacement is inferred. Small-loss recovery, analytic gain and evolved interventions remain supporting evidence, not the experimental headline.

Narrative points and conclusions

- Incumbent growth takes different spatial forms: most models widen; Goryachev grows taller and lowers available inactive substrate.
- Under the native stimulus-times-inactive channel, Goryachev has weaker size-dependent challenger activity; its L39 response is positive and later decays. WP has nearly size-invariant challenger height but a larger remaining incumbent domain at L39.
- After withdrawal WP39 retains two observed active domains while Goryachev39 restores the initial-side pattern over the finite follow-up. These are native response routes, not unique mechanistic classifications.

Proposed measurements and experimental follow-up

Activity peak/integrated increment inside the actual physical cue support, with native scale and a fixed within-model cue schedule.

Incumbent concentration, half-contrast extent/edge retreat and activity beyond the cue, followed after withdrawal.

Basal excess amount, connected width and peak concentration, preserving all97 native checkpoints and explicit reference normalization.

Scope, limitations and pending validation

Native stimulus channel, dose and timescale are not experimentally calibrated; no optical actuator or reporter model is selected.

Finite 13000-unit off observations do not establish asymptotic stability, site ancestry, whole-field replacement from a scalar gate, or a movement law.

Goryachev39 has four native sample centres in its incumbent core; the retained evidence is not a spatial-convergence demonstration.

Observed response signatures and organization do not uniquely identify molecular mechanisms or causally partition reversal resistance.

Figure 2. Distinct finite responses to local polarity challenges · continued

Sources and reproducibility

Methods, exact source arrays and checks

Native protocols and endpoint comparisons

Original unchanged organization artwork

Figure 3 review: constructing the activity vector · image 1 of 3

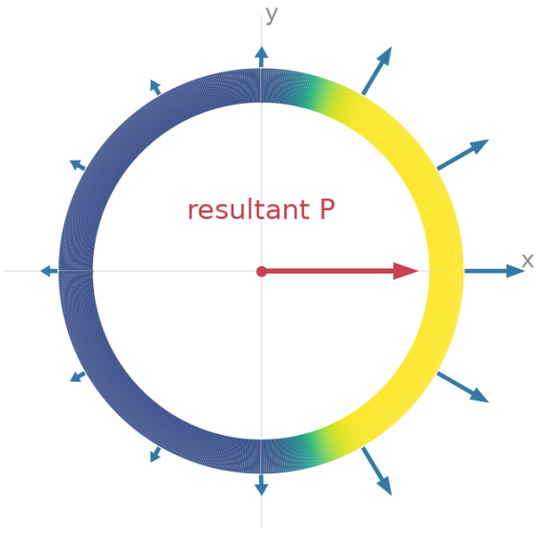
Two activity peaks can leave a weak resultant

Activity u is real; x + iy stores its weighted normal's two coordinates.

$$M_1 = \sum_j u_j \Delta s_j (\cos\theta_j + i\sin\theta_j), \qquad M_0 = \sum_j u_j \Delta s_j$$

$P = M_1/M_1(0)$ rotates and scales the initial resultant to 1.

A Initial basal activity



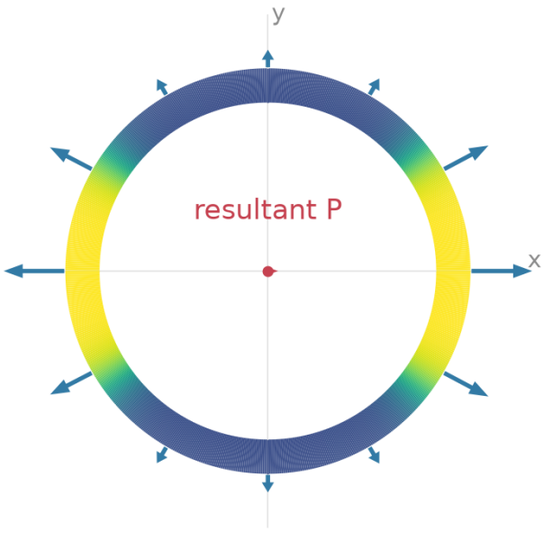
Before the antipodal cue

Total activity $M_0/M_0(0) = 1.0000$

Resultant magnitude $R = |P| = 1.0000$

Initial normalization sets P(0) = 1.

B Retained endpoint: opposing peaks



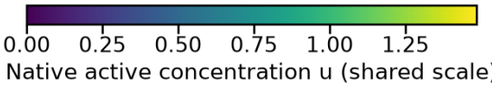
1000 cue units + 13,000 cue-free units

Total activity $M_0/M_0(0) = 0.9975$

Resultant magnitude $R = |P| = 0.0399$

Heading is not interpreted when R < 0.1.

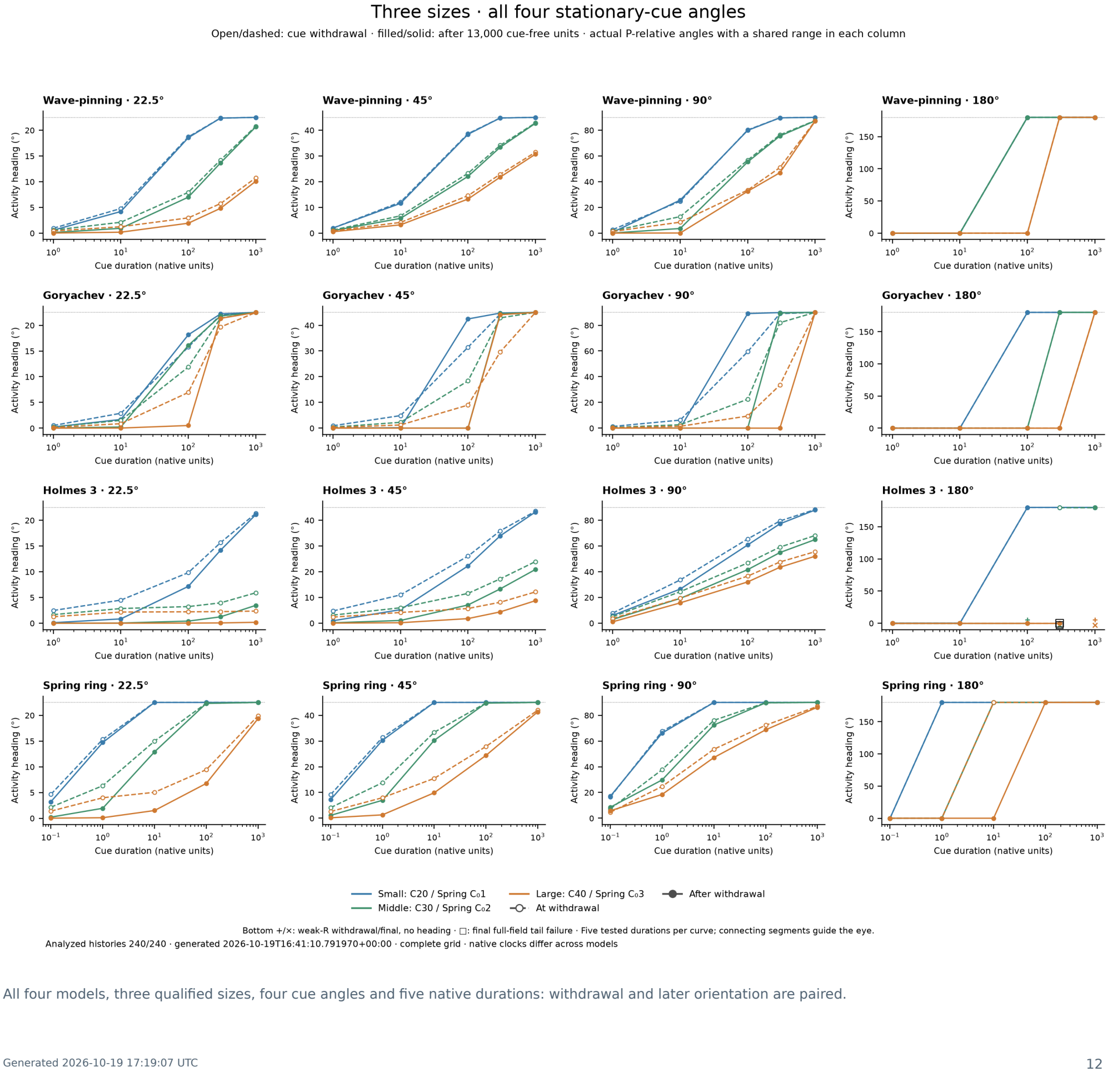
Blue arrows: exact sums of native cell contributions in 30° sectors.



All arrows use one vector scale.
Holmes 3 · fixed circumference C = 40.

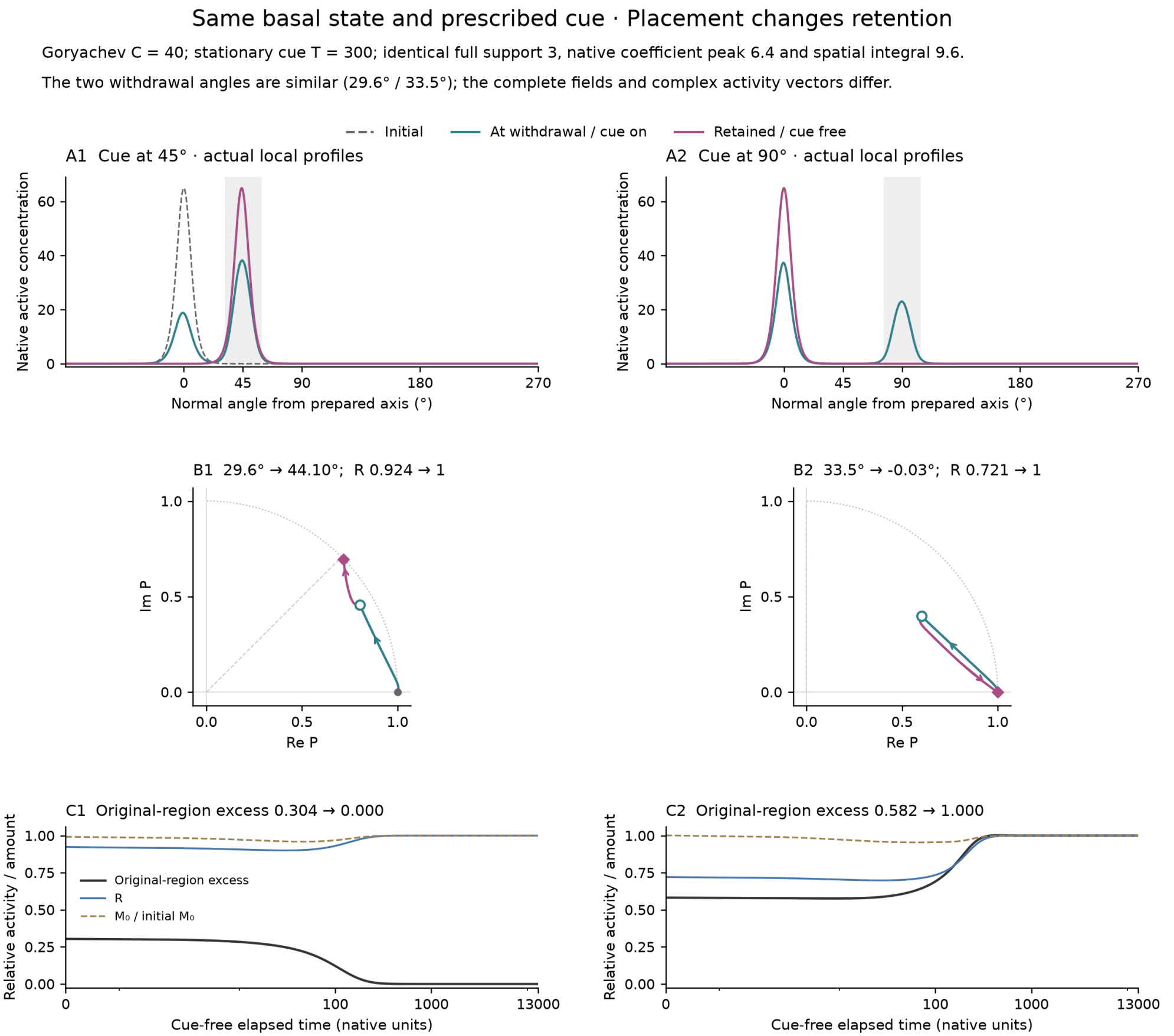
Measured Holmes profiles illustrate the physical vector sum and the distinction between resultant magnitude and total activity.

Figure 3 review: the complete stationary-cue experiment · image 2 of 3



All four models, three qualified sizes, four cue angles and five native durations: withdrawal and later orientation are paired.

Figure 3 review: a controlled placement contrast · image 3 of 3



A: full-ring profiles; grey band marks input support. B: equal scales; circle = withdrawal, diamond = +13,000. C: time is linear near zero, logarithmic later.

Only cue placement changes; withdrawal states differ. Both initial supports miss the half-contrast core, with different small basal tails.

Regional activity does not track site lineage. This selected contrast does not identify a general fate predictor, basin boundary or unique mechanism.

The same prescribed stationary cue is placed at two positions on the same Goryachev basal state; local profiles and complex vectors show the measured responses.

Figure 3. Local activity and retained orientation under stationary cues

One stationary-cue experiment links spatial activity to its global vector, then tests how cue placement, duration and model size affect response and retention. The following sheets are current evidence for choosing a cohesive main figure.

Detailed legend

Vector construction. Each boundary segment contributes its real active concentration multiplied by physical segment length and its outward unit normal. The complex first moment stores the summed x and y components. Dividing by its initial complex value rotates and scales the initial vector to one. Its angle describes activity orientation; its magnitude R describes the resultant relative to its initial magnitude. Total active amount M0 is measured separately. In the measured Holmes C40 antipodal example, final total activity is 0.99746 of initial while R is 0.03993: two opposing peaks nearly cancel. Blue arrows group exact native cell contributions into twelve sectors using the same vector scale as the red sum.

Experimental design. Every condition restarts an exact qualified basal state. Chemical circumferences are 20, 30 and 40; Spring resting circumferences are 1, 2 and 3. Cue centres lie at pi/8, pi/4, pi/2 or pi relative to the declared preparation axis. The frozen measured initial first-moment phase defines the small exact offset in plotted P coordinates; centres are never snapped to a different cell. Chemical cue durations are 1, 10, 100, 300 and 1000; Spring durations are 0.1, 1, 10, 100 and 1000. Each is followed by 13000 cue-free native time units.

Complete response map. Rows identify models, columns cue angles and colours the three sizes. Open/dashed curves show withdrawal; filled/solid curves show the later endpoint. Actual angles require R at least 0.1. Weak-resultant endpoints are explicitly marked without assigning them a heading. The curve joins guide the eye between five tested durations. All 240 conditions and all 125 initial cue/core overlaps are retained. The fixed physical full support is 3 for chemical models and 0.2 for Spring; native peaks are 0.1, 6.4, 1 and 4 for Wave-pinning, Goryachev, Holmes and Spring. Clocks and doses are not calibrated across models.

Controlled placement contrast. Goryachev C40 receives a 300-unit stationary cue at 45 or 90 degrees. Both conditions share the full basal state, kinetic parameters and cue shape, strength and duration. The cue arrays differ by an exact 100-cell cyclic shift; both have integral 9.6 and coefficient peak 6.4. These match the prescribed input coefficient; the native activation term also multiplies local inactive substrate, so equal activated biochemical amount is not imposed. Withdrawal headings are 29.59 and 33.48 degrees, but relative resultant magnitudes are 0.924 and 0.721 and both profiles contain two resolved peaks. After release, the 45-degree case progresses to 44.10 degrees while the 90-degree case returns to -0.028 degrees. The original material-region excess approaches 0 and 1, respectively. These are different complex vectors and local states at withdrawal.

Figure 3. Local activity and retained orientation under stationary cues · continued

Narrative points and conclusions

- Ring measurements distinguish graded retained reorientation from nearly cancelling activity. Across all 180 non-antipodal conditions, the final field is a shifted copy of its own basal pattern with one resolved activity region and recovered resultant magnitude. Intermediate angles therefore reflect a retained spatial shift rather than a weak final activity pattern.
- The finite-cue response depends on size within each native protocol. For a 90-degree cue of duration 100, Wave-pinning retains 79.86, 55.36 and 32.52 degrees across circumferences 20, 30 and 40; Goryachev retains 89.09, -0.014 and -0.003 degrees. The complete map preserves other angles and durations rather than reducing them to an arbitrary effective-duration threshold.
- Cue-on deflection and later orientation are distinct observations. The same-basal placement pair demonstrates how similar withdrawal angles can accompany different resultant magnitudes, local profiles and subsequent activity redistribution. This supports examining the activity vector and spatial pattern together; it does not identify a unique competition mechanism.

Proposed measurements and experimental follow-up

For every condition retain activity direction and resultant magnitude during the cue and release, total active amount, actual local profiles and the fixed original material-region excess. The rich tables include sampled minima/maxima, rebound, and first/retained threshold-entry brackets with finite-horizon censoring. Regional depletion can result from translation, redistribution or local chemistry and is not a site-ancestry measurement.

A boundary activity recording could supply the same weighted sum using measured boundary elements and their actual outward normals. The circle provides those normals exactly in the model. Relating this descriptor to cell movement would require an additional empirical measurement; movement is not inferred here.

Scope, limitations and pending validation

The size series keeps physical cue support and native parameters fixed. Physical cue separation, angular footprint, overlap, basal architecture and material totals consequently change with size. These experiments measure that declared protocol response rather than an isolated causal contribution of geometry.

The separate finite-tail field gate passes 228 conditions, is unavailable for 10 compact reused histories and fails for Holmes antipodal duration 300 at circumferences 30 and 40. Values and flags remain displayed. The second weak final resultant is Holmes C30/180 degrees/duration300 (R about 0.0997); the C40/180 degrees/duration1000 teaching

Figure 3. Local activity and retained orientation under stationary cues · continued

example has R about 0.0399. A finite release does not prove asymptotic stability. Repeated samples from one deterministic history are not independent replicates.

The figure composition remains under author discussion. These review sheets preserve the complete systematic evidence and a controlled illustrative pair; they are not an accepted final main-panel layout. The integrated outline assembles these provisional review sheets; the preceding prototype is preserved separately.

Model definitions and readouts

$$M_1(t) = \sum_j u_j(t)\,\Delta s_j(t)\,[\cos\theta_j + i\sin\theta_j] \tag{3.1}$$

The activity field u is real. The real and imaginary sums are Cartesian components of the physical activity vector. On each uniformly sampled circle, segment length is actual circumference divided by cell count, including Spring C(t).

$$P(t) = \frac{M_1(t)}{M_1(0)}, \qquad R(t) = |P(t)|, \qquad \phi(t) = \arg P(t) \tag{3.2}$$

The initial normalization is applied once; it aligns the initial axis and sets the initial magnitude to one. Heading is displayed only above the declared relative-resultant gate.

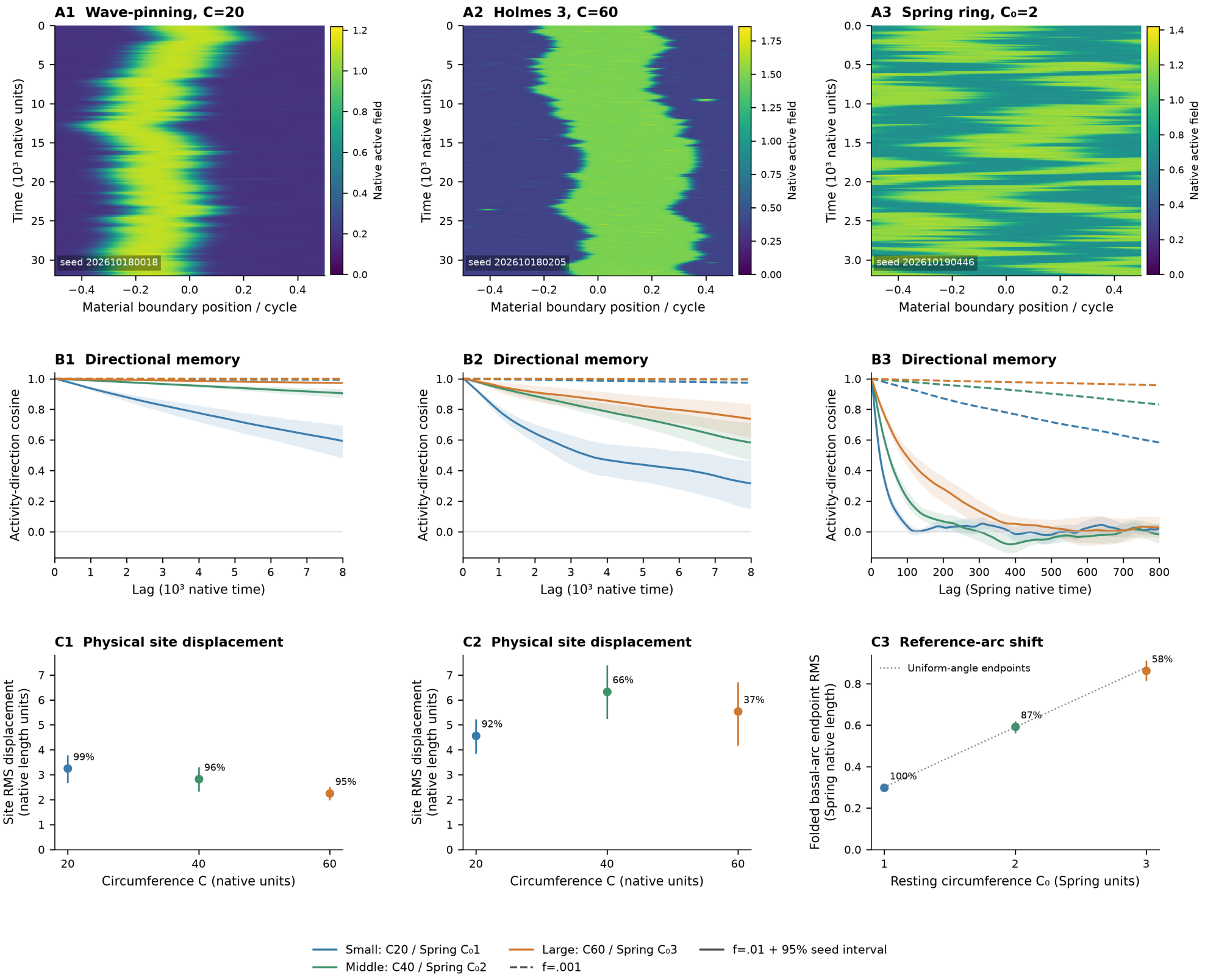
$$M_0 = \sum_j u_j \Delta s_j, \qquad q_1 = \frac{|M_1|}{M_0}, \qquad R(t) = \frac{M_0(t)}{M_0(0)}\frac{q_1(t)}{q_1(0)} \tag{3.3}$$

Relative resultant magnitude combines total activity and directional organization. It can exceed one and differs from current-mass-normalized coherence q1.

Sources and reproducibility

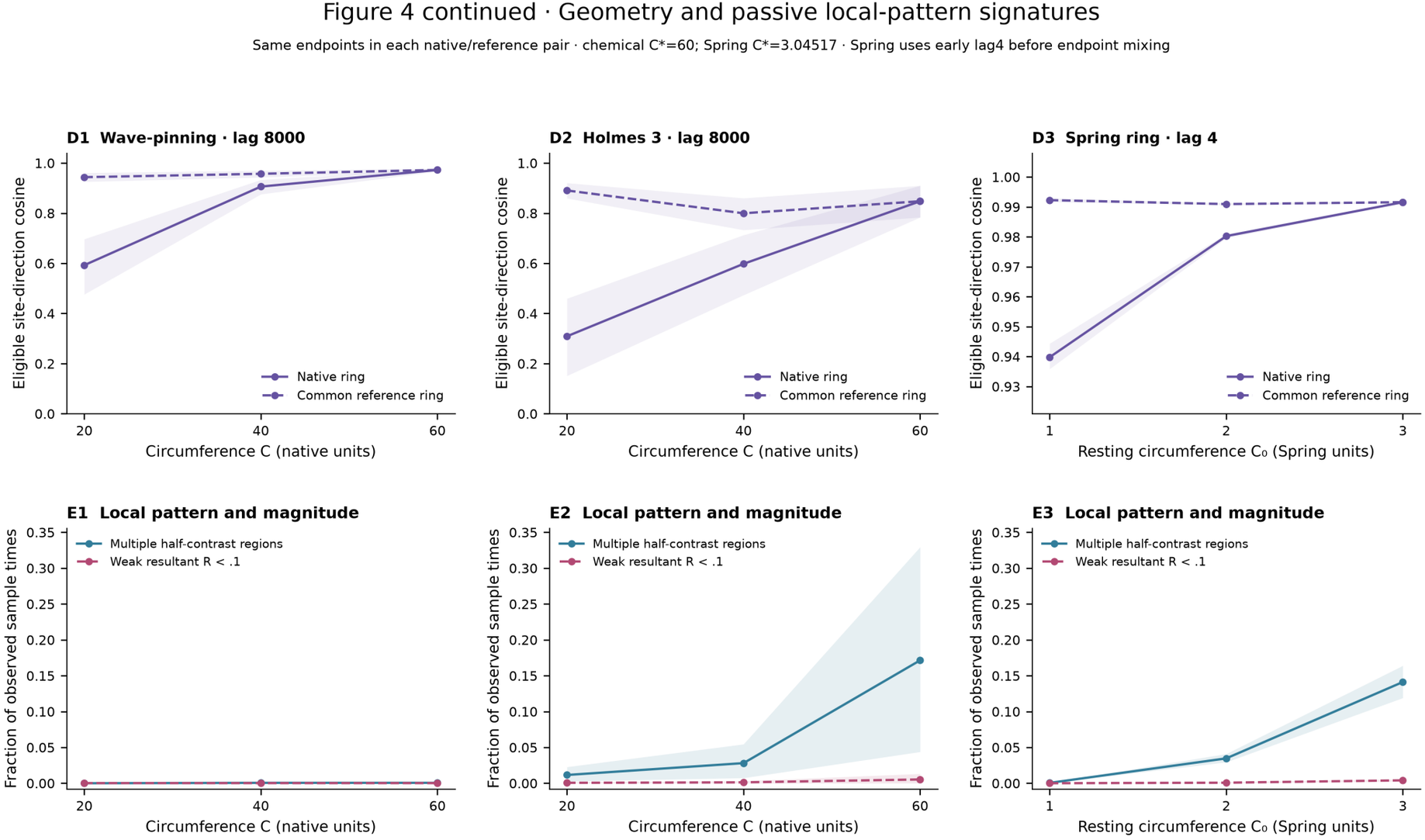
Authoritative complete 240-condition table
Initial profiles, cue footprints and overlaps
Regional crossing brackets and rebound flags
Native archive replay and complete-grid validation
Response-map evidence and enlarged supporting views
Exact physical vector contributions in the teaching example
Controlled placement-contrast evidence
Independent scientific review

Figure 4 · Directional memory and physical site dynamics · image 1 of 2



Native histories, activity-direction memory and physical/reference-arc site shifts.

Figure 4 · Directional memory and physical site dynamics · image 2 of 2



Paired geometric remapping and passive local-pattern/magnitude readouts.

Figure 4 · Directional memory and physical site dynamics

stochastic dynamics

Detailed legend

Figure 4. Directional memory needs a physical site readout. Memory panels B show both predeclared strengths: dashed f=.001, solid f=.01. Histories and site/pattern metrics use f=.01. Every noisy group retains all 12 independent seeds; shading and bars are pointwise 95% whole-seed intervals (B shades the stronger input). Native units differ across models.

A1–A3: native active-field histories from the predeclared condition medoids: Wave-pinning C20, Holmes C60 and Spring $C_0$2. Each model uses a zero-based native activity scale; boundary position is material position around the ring.

B1–B3: whole-window activity-direction cosine, conditional on both endpoint relative first-moment magnitudes R≥.1. The physical activity moment $M_1(t)$ is divided by its initial complex value. Activity direction is distinct from cell movement.

C1–C3: dominant half-contrast-site endpoint RMS shift at lag 8000 for chemical rings or 800 for Spring; percentages are eligible endpoint pairs. Spring uses basal-reference arc shift Cbasal×wrapped material-position change, with the dotted uniform-angle reference Cbasal/√12. Long-lag folded shifts can approach circular mixing and are not travel distances. Spring has its own native length axis.

D1–D3: native site-angle cosine and remapping of the same eligible shortest shifts to common circumference C*=60 for chemical rings or 3.04517 for Spring. Chemical panels use lag 8000; Spring uses early lag 4 before late endpoint mixing. Full lag curves and 4/50/800 comparisons remain supporting evidence. This is a geometric coordinate comparison, not a causal decomposition of global activity-vector memory.

E1–E3: observed time fractions with multiple current half-contrast activity regions and weak relative resultant R<.1. Region multiplicity does not establish functional competition. All 6401 scalar samples per seed enter each fraction. Supporting increments retain rare >π/4 Spring changes at cadence .5 and none in chemical traces at cadence 5, conditional on both endpoint R≥.1; these sampled diagnostics do not define an event class or exclude faster changes.

Complete Goryachev results remain supporting evidence: its high-input C20 mean control also loses polarity, precluding an intact-state wandering interpretation. Both declared strengths, all mean/no-input controls and medoids, magnitude and increment distributions, eligibility sensitivities and full/half-window analyses are retained. Methods define Spring's approved stretch/dilution/diffusion carryover, mean-force global-extension closure and finite transported-material noise law.

Narrative points and conclusions

Figure 4 · Directional memory and physical site dynamics · continued

- Angular-memory and physical site-shift measurements can show different size trends. Spring makes the forcing dependence explicit: weak-input lag-800 direction cosine rises .584/.832/.958 across sizes despite nonmonotone basal-arc RMS, whereas stronger input gives .019/−.017/.028 with substantial mean relative first-moment magnitude remaining and late folded shifts near the uniform-angle reference. All three main models retain their native units and controls.
- Pairwise coordinate remapping exposes the geometric part of site-angle retention, while local-region and magnitude readouts distinguish observable organizations without classifying abrupt events or inferring competition.

Proposed measurements and experimental follow-up

Activity-direction correlation and full-vector memory

Physical/reference-arc site RMS with endpoint coverage

Native/common-reference site-angle retention on identical pairs

Local-region and weak-resultant occupancy

Gated increment distributions and full/half-window sensitivity

Scope, limitations and pending validation

Native units are not calibrated across models.

Activity direction does not assign movement.

Endpoint pairs do not establish site ancestry.

Spring reference-arc shifts separate material change from uniform stretch.

Remapping is a coordinate comparison, not a causal decomposition.

Finite-knot/noise-law and finite-window limitations remain explicit.

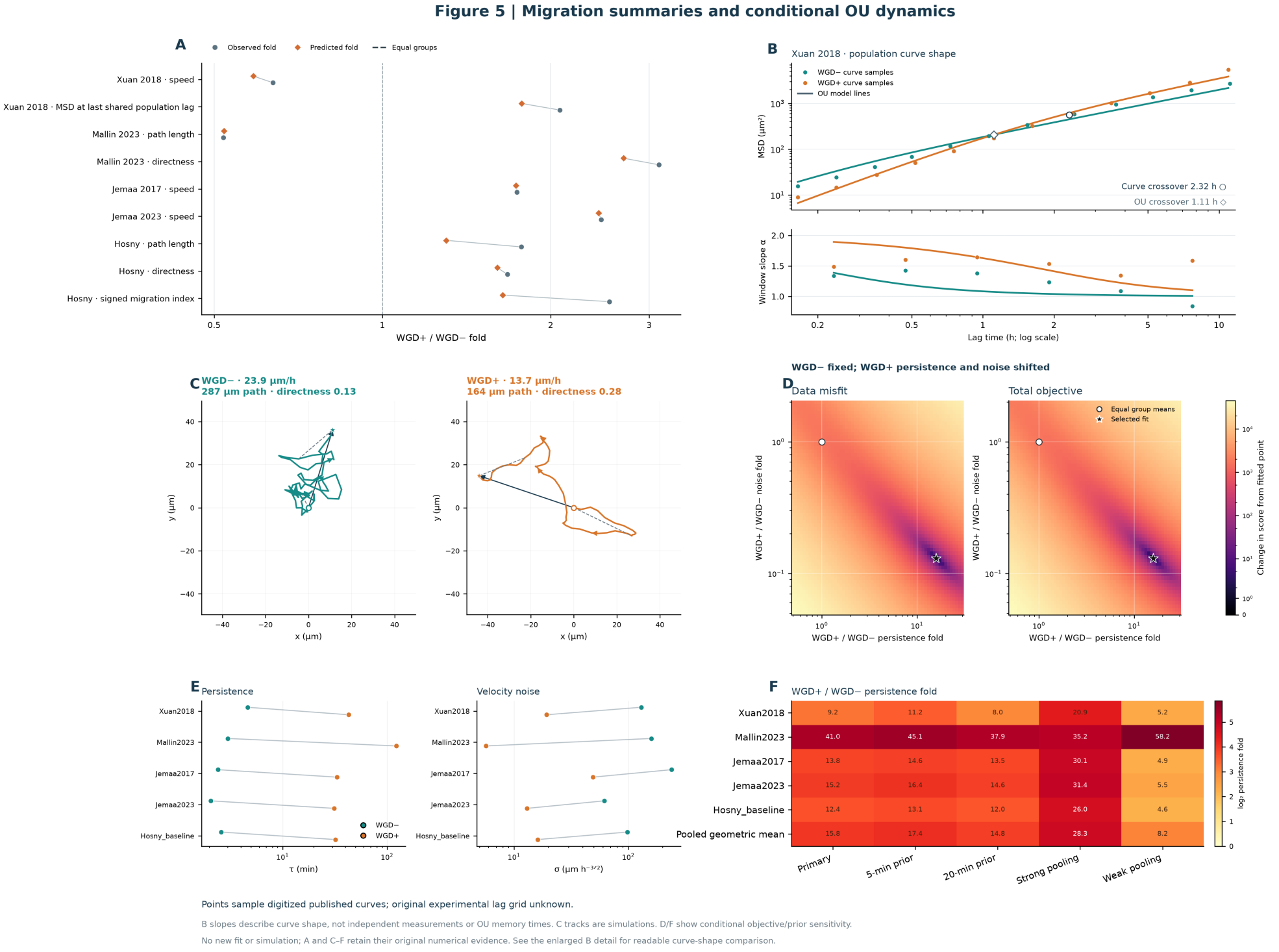
Sources and reproducibility

Original chemical-ring stochastic report
Original noise design
Complete Goryachev/chemical interim figure
Frozen Spring main design
Paired finite-law estimator sensitivity
Spring full analysis audit
Spring centroid/half-window/CDF/forcing audit
Spring full seed coverage
Spring methods and limits

Figure 4 · Directional memory and physical site dynamics · continued

Final four-model quantitative evidence
Spring raw full/half-window memory
Spring site and stretch metrics
Spring first/last-half site metrics
Spring magnitude-gated increment distributions
Spring actual forcing audits
Spring early/late endpoint summary
Independent scientific review
Independent requirement and evidence audit (assembly snapshot)
Spring active amount and spatial-balance methods
Spring scalar amount/profile audit
Spring active amount and balance condition summaries
All81 Spring scalar case exports

Figure 5. Migration observations and conditional OU interpretation · image 1 of 2



Points sample digitized published curves; original experimental lag grid unknown.

B slopes describe curve shape, not independent measurements or OU memory times. C tracks are simulations. D/F show conditional objective/prior sensitivity.

No new fit or simulation; A and C–F retain their original numerical evidence. See the enlarged B detail for readable curve-shape comparison.

Reassembled Figure 5: original A and C–F numerical evidence, revised MSD/curve-shape panel B.

Figure 5. Migration observations and conditional OU interpretation · image 2 of 2

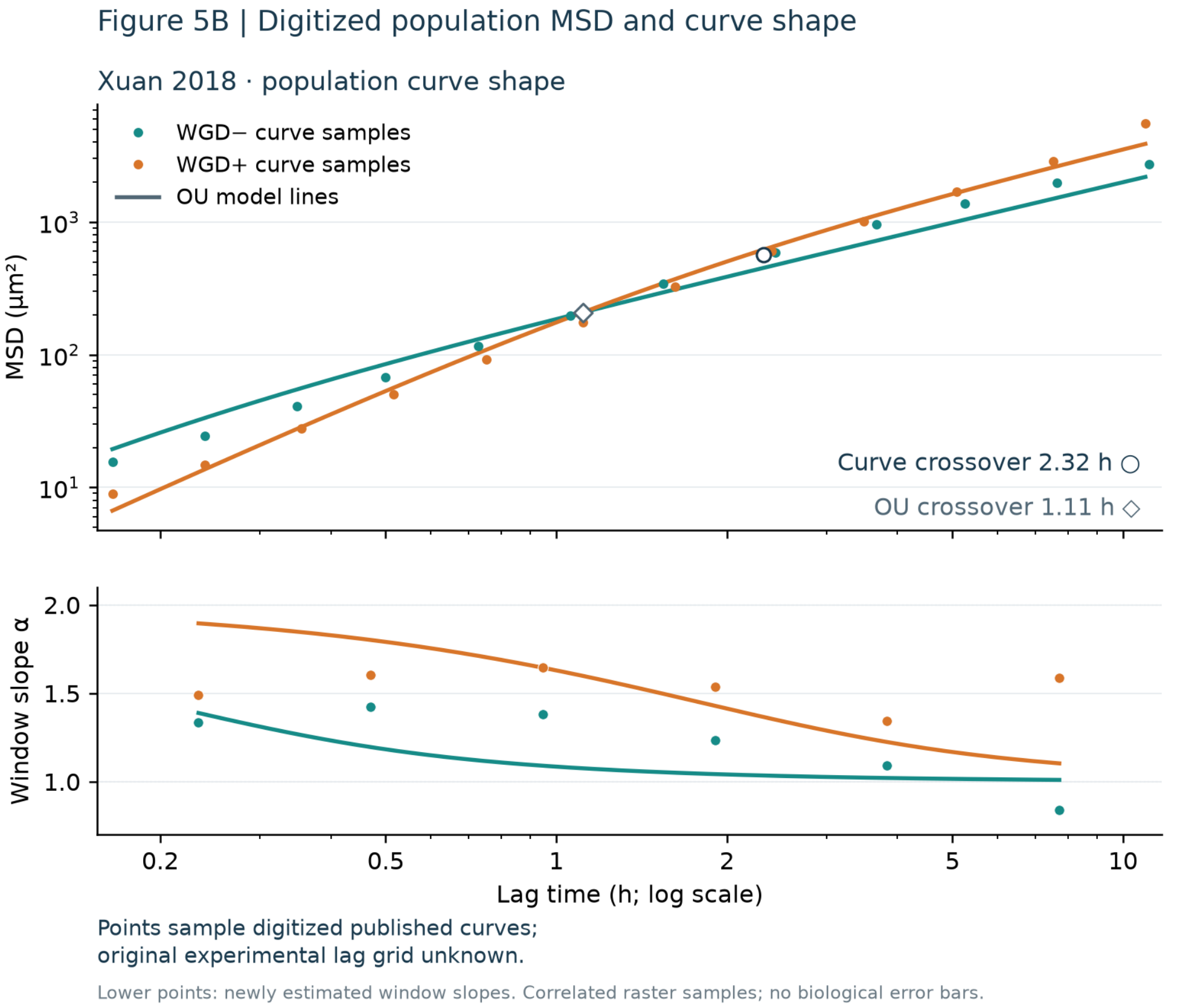


Figure 5B continued: enlarged detail with explicit raster-sample caveat.

Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation

Existing empirical support, with observed migration summaries kept separate from fitted velocity-memory parameters.

Detailed legend

A, Published/digitized enlarged or polyploid-group to reference-group ratios for nine migration summaries, alongside primary Ornstein–Uhlenbeck (OU) fit predictions. Xuan contributes sampled mean speed and population mean-squared displacement (MSD) at the last shared lag, approximately 10.95 h; Hosny summaries are group medians. Mallin central-bar statistic is not specified by the source and retains the historical mean approximation. These distinct source summaries are not interchangeable measures of persistence.

B, Upper: twelve chosen display samples from each digitized Xuan population MSD curve (small points) and the unchanged fitted analytical OU expectation (solid lines), on logarithmic lag and MSD axes. These are samples of the published raster curves, not recovered experimental lag positions or independent cell observations; the original experimental lag grid is unknown. The open circle identifies the crossover of the full digitized traces at approximately 2.32 h, computed by linear interpolation in native lag and MSD. The open diamond identifies the different fitted OU crossover at approximately 1.11 h. Lower: descriptive slopes of log(MSD) versus log(lag) fitted in six equal log-lag windows of the full digitized curves (points), and matching-window moving OU slopes (solid lines). Each window spans approximately 0.304 decades; both observed and model slopes use the same full window support. Observed raster columns remain correlated and give no biological sample size or confidence interval. These slopes are newly calculated curve-shape summaries, not author-reported local exponents or fitted OU memory times. A constant power-law exponent would appear horizontal; reference values 1 and 2 indicate diffusive and straight-line scaling. The model does not reproduce the complete observed curvature or the observed crossover. The analytical fit, all fitting targets and numerical inputs to A and C–F are unchanged.

C, Representative Xuan primary-fit model-generated tracks, with sampled speed, total path length and directness (net displacement divided by path length). Both groups use identical spatial scales. The tracks are simulations, not published individual-cell trajectories; a slower sampled trajectory can nevertheless have greater endpoint directness. Dashed chords illustrate the contribution of finite-time displacements.

D, With reference-group parameters fixed, vary enlarged-group persistence and velocity-noise ratios. Separate surfaces show change in the descriptive data-misfit score and in the total regularized objective; the selected fit and equal-group reference are marked. The heuristic score is not a calibrated likelihood, and the surfaces are not confidence regions.

Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation · continued

E, Primary paired persistence and velocity-noise estimates for Xuan 2018, Mallin 2023, Jemaà 2017, Jemaà 2023 and the Hosny hypoxic baseline. F, Enlarged/reference persistence folds under the primary 10-minute prior, 5- and 20-minute alternatives and stronger/weaker pooling, with the pooled geometric mean. Priors and pooling define conditional sensitivity scenarios, not uncertainty intervals. The Jemaà comparisons constrain speed alone, leaving persistence/noise tradeoffs dependent on hierarchy and priors.

The fitted population contrasts span stable tetraploids, giant polyploids and treatment-induced polyaneuploid cancer cells (PACCs). The inherited WGD+/WGD− labels in the artwork are comparison shorthand and do not establish matched WGD-only interventions in every study. Hosny observations derive from the recovered preprint; equivalence to the journal version remains unresolved.

Narrative points and conclusions

- Xuan combines lower sampled speed (0.64-fold) with greater late-lag population MSD (2.08-fold); Mallin combines shorter path length (0.52-fold) with greater directness (3.12-fold). Persistence-related behaviour therefore deserves analysis separately from speed.
- Hosny baseline PACCs have greater median path length (1.77-fold) and directness (1.67-fold). Signed migration index is a separate directional summary.
- All five primary OU comparisons infer longer velocity memory. The pooled primary persistence fold is about 15.8, and remains above one across the declared prior/pooling scenarios; this is a conditional model result.
- The fit does not reproduce the complete Xuan curve shape within its raster envelope, and data-only sensitivity has rank 18 of 22. Optimization completion does not establish model adequacy or uniquely identified persistence times.
- The observed Xuan curves cross near 2.32 h while the unchanged fitted OU curves cross near 1.11 h. Matched-window slopes make the remaining curve-shape discrepancy visible; fitted persistence is a conditional summary.
- Lower sampled speed can coexist with greater late-lag displacement in the enlarged-group comparison. Its causal connection to polarity mechanisms requires matched activity and migration measurements.

Proposed measurements and experimental follow-up

The new WGD assay should measure matched trajectories directly, with declared sampling and observation duration, and assess speed, displacement, directness and lag-dependent MSD before considering fitted persistence. An activity-polarity memory

Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation · continued

measurement would complement these quantities without equating activity and velocity.

Scope, limitations and pending validation

Neither these heterogeneous group comparisons nor the OU fits establish that enlargement alone causes persistence. Cell identity, induction/treatment and acquisition differences require explicit treatment in the new experiments.

The requested observed-point/model-line and curve-shape display is implemented. The original experimental lag grid remains unknown; neither display samples nor raster-derived slopes supply independent biological replicates. The observed/model crossover discrepancy remains visible.

Sources and reproducibility

Refined OU analysis
Original panel legend
Observed folds
Fit results and adequacy
Hosny sensitivity supplement
Revised display, exact preserved inputs and direct-user feedback
Migration display numerical and preservation checks

Figure 6. RESERVED: new WGD motility results



Explicit placeholder. Every panel is pending acquisition, analysis and author review; no mock observations are displayed.

Figure 6. RESERVED: new WGD motility-assay results

A substantial new experimental results section and main figure. No observations are available; the panel layout is a proposal, not an experimental result.

Detailed legend

A, Proposed experimental context: the SUM-159 chemically induced WGD+/− pair is the agreed starting pilot with the Speros collaboration. Imaging should first be optimized for that pair before expansion to additional cell lines. Ploidy/group identity, physical size and morphology need to be documented so the intended comparison is clear.

B, Reserved for representative experimental fields and trajectories under the selected acquisition conditions. The eventual panel should state sampling interval, usable duration, spatial scale, track eligibility and exclusions. Number of independent preparations and biological replicates remains to be specified.

C, Reserved for empirical motility summaries: sampled speed, accumulated path length, endpoint displacement/directness and lag-dependent MSD. The final readouts, follow-up window and analysis plan are to be settled before interpreting the data; this placeholder does not imply that all listed measurements will be technically feasible.

D, Reserved for the measured relationship between WGD-associated enlargement and migration behaviour, including size distributions and controls needed to interpret the group contrast. Any activity/protrusion measurements arising from the mechanism proposals require a separately agreed assay and should be distinguished from the basic motility pilot. Statistical models, uncertainty reporting and sample-size targets remain open.

Narrative points and conclusions

- This major results section will test whether the planned paired experimental system exhibits the migration behaviour motivating the theoretical work.
- Its conclusions must follow the acquired observations, including null or opposing results. Simulation output and literature fits cannot fill this gap.
- Agreement between increased polarity resistance or activity memory and migration persistence would support the proposed connection; causal attribution requires appropriate perturbations and controls.

Proposed measurements and experimental follow-up

Pending decisions include plating density, acquisition duration and cadence, field/condition throughput, tracking and exclusion rules, replicate structure, confirmation of WGD status, size quantification and the statistical analysis. The known throughput versus spatial/temporal resolution trade-off should be resolved by the pilot.

Figure 6. RESERVED: new WGD motility-assay results · continued

Mechanism validation can subsequently examine incumbent extent, early challenger growth, local loss/rebound, site fates and global activity-polarity correlation under characterized perturbations. These activity experiments are proposals alongside the theoretical mechanisms and are separate from the reserved motility-results contribution.

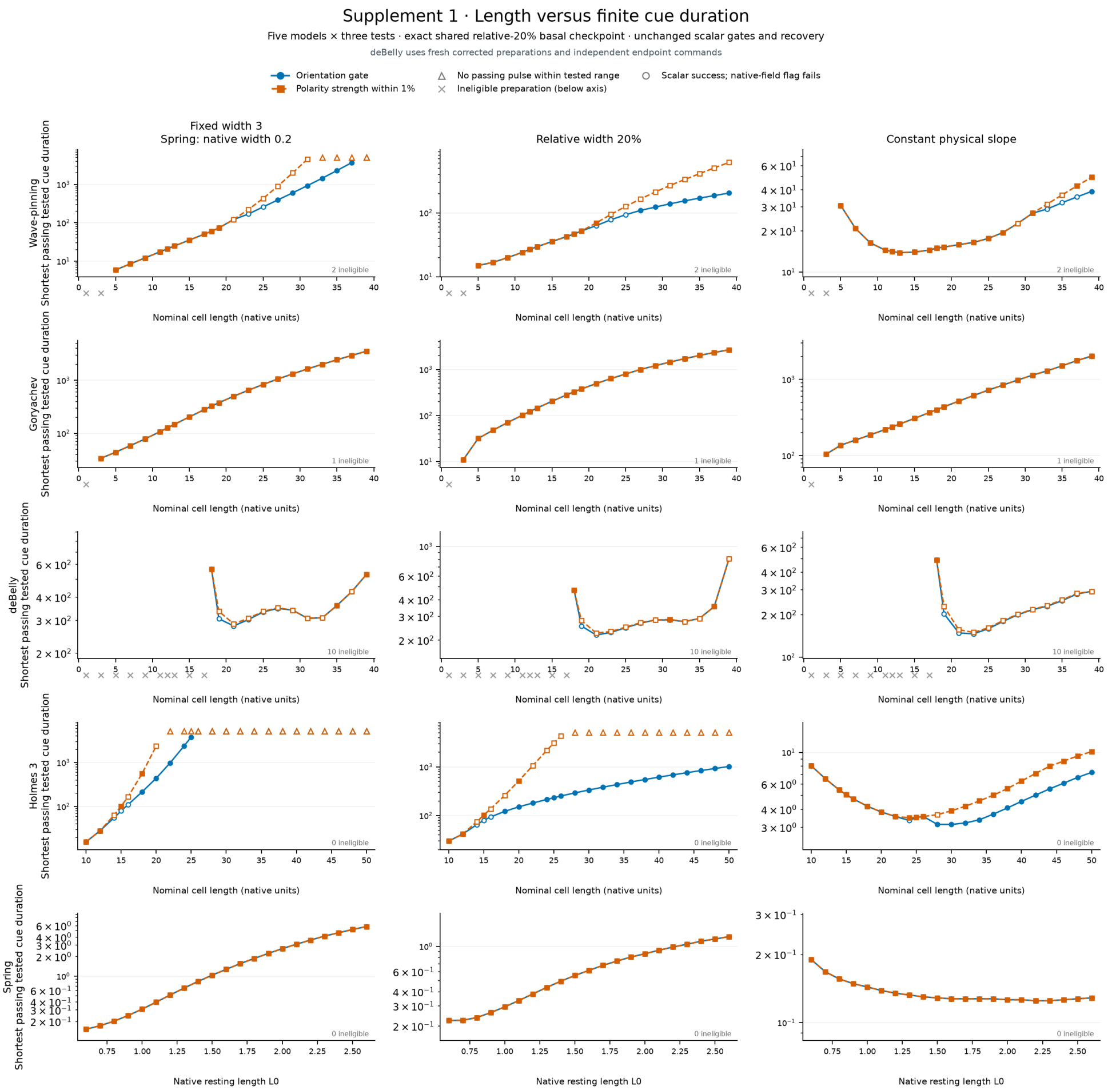
Scope, limitations and pending validation

No pilot acquisition, sample size, effect estimate, significance claim or experimental outcome is supplied by the available evidence.

Sources and reproducibility

Author manuscript vision and supplied collaboration context

S1. Full cue-duration response curves



Both patches use a constant peak across all lengths; Goryachev K=64, peak=6.4. Gradients retain one constant physical slope per model.
Each positive duration restarts the original checkpoint. Full strength adds the 1% reversed initial-polarity band to the orientation gate.
Open markers retain failing field-tail flags. Censoring marks explicit tested limits; missing endpoints break lines. Native times and scales differ.
Data generated 2026-10-08T19:26:33.200718+00:00; producer finished. Every imported and fresh trial passed independent native-array audit.

Canonical corrected five-model duration curves; source working number Supplement 1.

S1. Full cue-duration response curves

Essential curve context for the main comparison heatmap.

Detailed legend

Five model rows and three finite-cue columns retain the complete sampled length dependence. Coloured curves distinguish orientation from strength replacement. Open success markers retain scalar successes that fail the separate native-field tail flag; triangles indicate no passing pulse within the tested range; crosses indicate ineligible preparations. Missing values are not zero response times. Each duration excludes preparation and the subsequent 10,000-unit basal recovery. Spring uses native length and time. See Figure 1 for exact gates and stimulus conventions.

Narrative points and conclusions

- Corrected deBelly is nonmonotone across all three geometries. The positive rank coefficients summarize association rather than its full curve shape.
- Constant-slope Spring shows the opposite association to its patch responses. Holmes orientation under that cue varies with little monotone association. Censoring and native-field flags restrict interpretation even where rank correlation is strong.

Sources and reproducibility

Current figure legend and gate definitions

Supplementary material and supporting analyses

S2. Numerical qualification and preparation provenance

Exact common starting states, duration-boundary audits, corrected deBelly endpoint controls and selected mesh diagnostics; source histories remain distinguishable from compact committed summaries.

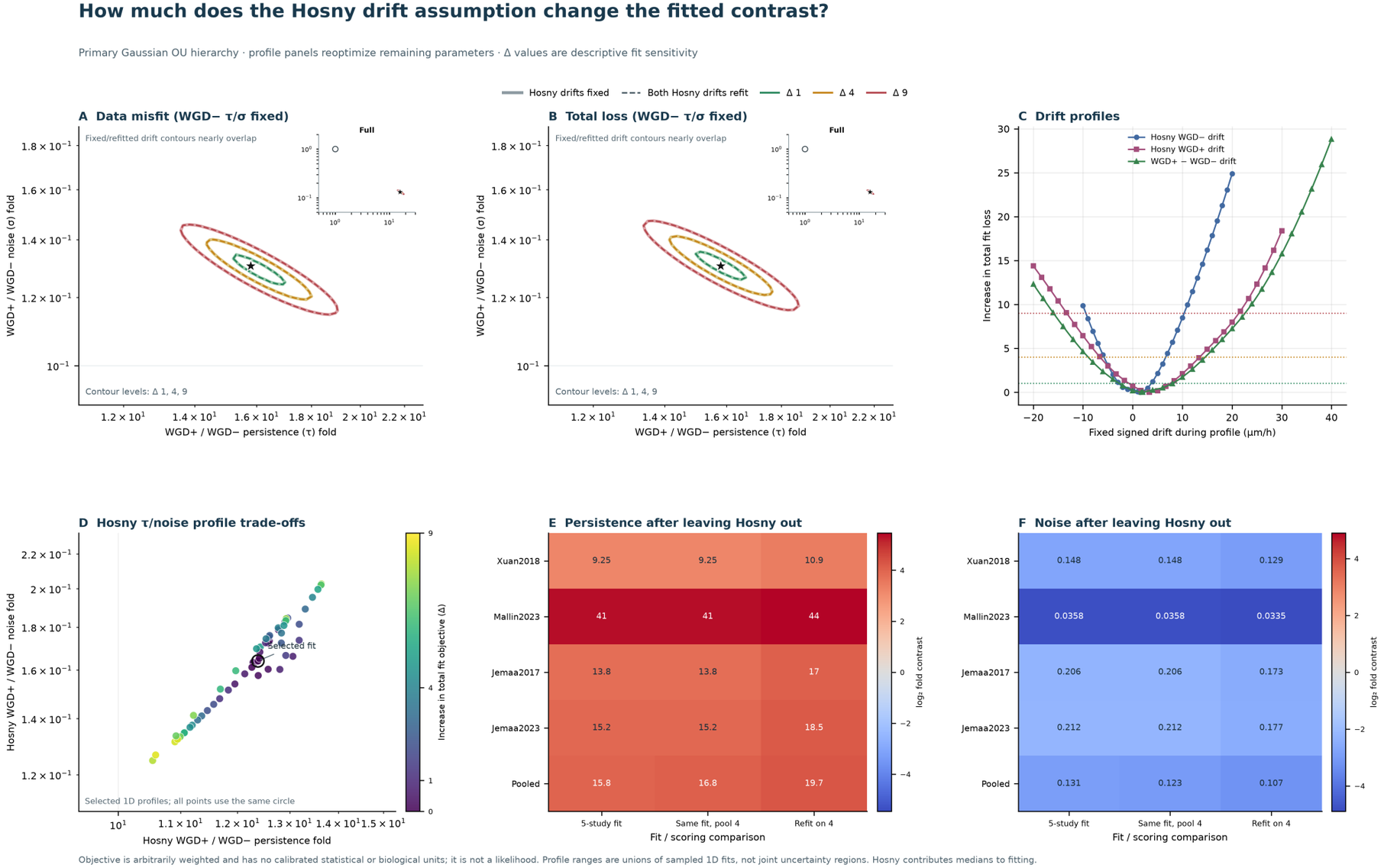
polarity1d/manuscript/THREE_TESTS.md

S3. Mechanism and stochastic sensitivity

Complete gain/recovery grids, clock qualification, finite-cue definitions, outcome tables, site/vector diagnostics and numerical/noise controls accompany the mechanism checkpoint. Revised Figure 4 also displays both strengths, complete half-window diagnostics, physical-site eligibility and matched C20/C60 native examples; compact source tables support portable rerendering.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/VISION_CHECKPOINT_REQUIREMENTS.md

S4. Migration-model adequacy and Hosny sensitivity



Existing Hosny sensitivity supplement from the refined OU analysis.

S4. Migration-model adequacy and Hosny sensitivity

Supporting conditional sensitivity analysis; it is not an independent biological replication.

Detailed legend

The retained supplement explores signed-drift and leave-Hosny-out sensitivity within the declared OU hierarchy. Its fitted changes and descriptive-score profiles are conditional on the model and supplied summaries, not confidence intervals. Data-only identifiability, Xuan population-curve residuals and prior/pooling assumptions remain part of the adequacy assessment. Representative movie diagnostics are separate: acquisition timing is uncertain and nuclear appearance does not establish ploidy.

Sources and reproducibility

Original Hosny supplement caption

Sensitivity methods and results

Gain and small-loss recovery: supporting evidence

Detailed legend

Calculated native input gain is separate from observed post-cue activity. The small-loss recovery contrast persists through stored finite trajectories but weakens with time; actual damage is only about0.22–0.285% of core activity, with no reporter calibration.

Scope, limitations and pending validation

Small-loss assays are not the finite optical challenge or a universal stability measure.

Sources and reproducibility

Finite-time and amplitude audit

All97 gain/recovery values

Selected native pulse and causal controls

Detailed legend

WP L15 T20/T60 starts from the same basal state and illustrates withdrawal-dependent domain survival. Evolved interventions have endpoint- and cue-specific limits.

Scope, limitations and pending validation

WP4× mobility passes orientation only at one dose; both conditions fail the stronger full-polarity-strength gate. Mobility is changed during cue and recovery.

Sources and reproducibility

Retained WP short/long pulse artwork

Endpoint-qualified controls

Supplementary material and supporting analyses

Stationary-cue endpoint-vector alternative

An alternative view displays all 240 measured endpoint vectors on equal coordinate scales, with weak-resultant, overlap and finite-tail flags retained. It is available for discussion without replacing the complete withdrawal/retention map in the current review.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure3/standard_angles_sizes/vector_grid/README.md

Spring: every native medoid

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_all_histories.png

Spring magnitude, geometry and controls

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_magnitude_geometry_controls.png

Spring full/half-window memory

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_memory_windows.png

Spring increment tails

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_increment_tails.png

Spring short/long-lag geometry and circular mixing

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_endpoint_lag_sensitivity.png

Spring largest sampled increment native context

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_large_increment_context.png

All chemical models including Goryachev

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_stochastic_revision.png

Chemical memory: both strengths

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_both_strengths_memory.png

Chemical magnitude and controls

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_magnitude_controls.png

Supplementary material and supporting analyses · continued

Chemical increment tails

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_increment_tails.png

Chemical site-gate sensitivity

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_site_gate_sensitivity.png

Original chemical medoids, all18 conditions

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/vision_noise_20261017/vision_noise_selected_histories.png

Goryachev lowest-resultant diagnostic

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/vision_noise_20261017/vision_noise_lowest_resultant_diagnostic.png

Spring active amount and spatial first harmonic

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_amount_spatial_balance.png

Preserved complete mechanism and assay checkpoint

The preceding outline retains all-five mechanism-to-measurement sheets, ring gain/recovery and finite-cue maps, and noise magnitude, controls and full/first/last-window diagnostics. Its numerical evidence is preserved; the revised main-figure claims are stated in this edition.

manuscript_outline/outline_v2.pdf

Prior all-five mechanism evidence crosswalk

Prior complete ring and noise checkpoint

Prior physical motion, magnitude and coverage evidence

References and evidence provenance

Reviewed primary literature and extraction map
Scientific scope and author direction
Direct-user preference evidence and foundational figure audit
Mechanism checkpoint intake and live-producer evidence
Source checkout, evidence selection and reproduction
Independent critical assessment and repair decisions
Portable original/renumbered mechanism replay and source closure
Second-edition integration decisions and validation scope
Figures 2–4 revision specification